

1. PLANT SCIENCE

Introduction to plant science, cell structure, and classification of plant kingdom

Q.1 Which of the following is not a characteristic of Angiosperm Plant Cells?

- (A) Cell wall (B) Chloroplasts
(C) Centrioles (D) Large central vacuole

Ans : (C) [FSO 2026]

Q.2 Which of the following is not a distinguishing feature of plant cells ?

- (A) Cell wall made of cellulose
(B) Presence of chloroplasts
(C) Lysosomes absent
(D) Presence of centrioles

Ans : (D) [FSO 2026]

Principles of plant physiology with reference to plant nutrition, absorption, transactions and metabolism of nutrients

Q.3 The final electron acceptor in the electron transport chain of respiration is

- (A) NAD⁺ (B) FAD
(C) Oxygen (D) Carbon dioxide

Ans : (C) [FSO 2026]

Q.4 In C₃ plants, CO₂ fixation initially occurs in

- (A) Mesophyll cells (via PEP carboxylase)
(B) Bundle sheath cells
(C) Stroma
(D) Cytoplasm

Ans : (A) [FSO 2026]

Q.5 Grass is rich in

- (A) Proteins (B) Starch
(C) Cellulose (D) Amino acids

Ans : (C) [FSO 2026]

Diagnosis of nutrient deficiencies and their amelioration

Q.6 Deficiency symptoms of an element tend to appear first in young leaves. It indicates that the element is relatively immobile. Which one of the following elemental deficiencies would show such symptoms?

- (A) Sulphur (B) Magnesium
(C) Nitrogen (D) Potassium

Ans : (A) [FSO 2026]

Q.7 Yellowing of older leaves while young leaves remain green indicates deficiency of

- (A) Nitrogen (B) Iron
(C) Calcium (D) Boron

Ans : (A) [FSO 2026]

Q.8 Dying back phenomenon is not observed in one of the following species.

- (A) *Tectona grandis* (B)
(C) (D)

Ans : 2 — *Tectona grandis* [FSO 2020]

Photosynthesis and respiration, growth and development, auxins and hormones in plant growth

Q.9 Plant hormone Indole-3-Butyric Acid (IBA) is used to

- (A) Prevent diseases (B) Enhance root initiation
(C) Reduce transpiration (D) Increase flowering

Ans : (B) [FSO 2026]

Q.10 Flowering of trees at an early age is known as

- (A) Precocious flowering (B)
(C) (D)

Ans : 4 — Precocious flowering [FSO 2020]

Significance of D.N.A and R.N.A in the cytology

Q.11 The function of mRNA is to

- (A) Carry amino acids
(B) Transcribe genetic information
(C) Translate genetic code into proteins
(D) Store genetic information

Ans : (C) [FSO 2026]

Q.12 The semiconservative model of DNA replication implies

- (A) Both strands are newly synthesized
(B) Each daughter DNA has one old and one new strand
(C) Only the leading strand replicates
(D) DNA replicates only in cytoplasm

Ans : (B) [FSO 2026]

Q.13 The DNA found in chromosomes is _____

- (A) Unique DNA and repetitive DNA
(B)
(C)
(D)

Ans : 3 — Unique DNA and repetitive DNA [FSO 2020]

Q.14 _____ is the study of structure & behaviour of chromosomes, their other properties and function in cells.

- (A) Cytogenetics (B)
(C) (D)

Ans : 2 — Cytogenetics [FSO 2020]

Vegetative, asexual and sexual methods of reproduction, Pollination and fertilization, Sexual incompatibility

Q.15 Double fertilization in angiosperms involves

- (A) Two sperm nuclei fusing with the egg and polar nuclei
(B) Fusion of two eggs with one sperm
(C) Fusion of pollen tube with ovule
(D) Formation of two zygotes

Ans : (A) [FSO 2026]

Q.16 Windpollinating (anemophilous) plants typically exhibit

- (A) Sticky pollen
(B) Small, inconspicuous flowers
(C) Large, brightly colored petals
(D) Heavy nectar production

Ans : (B) [FSO 2026]

Q.17 Which one of the following is a bad coppice?

- (A) Casuarina (B)
 equisetifolia
(C) (D)

Ans : 2 — Casuarina equisetifolia [FSO 2020]

Q.18 Lignotubers are underground swellings found in most species of

- (A) Eucalyptus (B)
(C) (D)

Ans : 4 — Eucalyptus [FSO 2020]

Q.19 Cross pollination between flowers borne on two different individual plants of the same species is known as

- (A) Xenogamy (B)
(C) (D)

Ans : 3 — Xenogamy [FSO 2020]

Q.20 _____ are groups of plants that are identical to their one parent and can only be propagated asexually.

- (A) Clones (B)
(C) (D)

Ans : 1 — Clones [FSO 2020]

Principles and classification of plant propagation methods

Q.21 Which of the following best defines regenerate?
(I) To rotate crops in different plots

- (A) To introduce exotic species into a region
(B) To renew a forest crop by natural or artificial means
(C) To remove old trees from the forest
(D)

Ans : (C) [FSO 2026]

Q.22 Which of the following is not a Hetero-vegetative method of plant propagation?

- (A) Layering (B)
(C) (D)

Ans : 3 — Layering [FSO 2020]

Q.23 Saddle, Cleft, Veneer, Side, Bridge, Bark, Root are various kinds of methods of vegetative propagation.

- (A) Grafting (B)
(C) (D)

Ans : 1 — Grafting [FSO 2020]

Q.24 What is a tree that has been recommended for production or breeding orchard use following grading, called?

- (A) Plus tree (B)
(C) (D)

Ans : 2 — Plus tree [FSO 2020]

Sexual propagation and its importance and the factors affecting germination and pre-germination treatments

Q.25 Seed treatment by which the hard seed coat is ruptured or weakened, mechanically or chemically is called

- (A) Scarification (B)
(C) (D)

Ans : 2 — Scarification [FSO 2020]

Q.26 Alternate wetting and drying is the seed treatment followed for _____ tree seed

- (A) *Tectona grandis* (B)
(C) (D)

Ans : 4 — *Tectona grandis* [FSO 2020]

Development, structure, dormancy and germination of seed

Q.27 Seed dormancy caused by a hard seed coat is best overcome by

- (A) Cold stratification (B) Scarification
(C) Hormonal inhibition (D) Water stress

Ans : (B) [FSO 2026]

Q.28 The type of germination of seed where cotyledons emerge out of the soil is called germination.

- (A) Epigeal (B)
(C) (D)

Ans : 3 — Epigeal [FSO 2020]

Q.29 Which of the following condition(s) is / are correct?

- (A) All productive soils are fertile.
(B)
(C)
(D)

Ans : 3 — All productive soils are fertile. [FSO 2020]

Q.30 Tree which do not form part of the upper most leaf canopy, but the leading shoots of which are not definitely over topped by the neighbouring trees are called as _____

- (A) Dominated trees (B)
(C) (D)

Ans : 2 — Dominated trees [FSO 2020]

Seed production and seed collection techniques in forestry crops

Q.31 The best time to collect seeds is when they are

- (A) Immature
(B) Fully mature but not dispersed
(C) Rotten
(D) Infested with pests

Ans : (B) [FSO 2026]

Q.32 A common technique for collecting seeds from tall forest trees is

- (A) Drilling (B) Climbing and pole-pruning
(C) Budding (D) Micropropagation

Ans : (B) [FSO 2026]

Q.33 Certified seed is the progeny of

- (A) Foundation seed (B)
(C) (D)

Ans : 1 — Foundation seed [FSO 2020]

Plant Diseases - factors affecting infections

Q.34 Which condition favors fungal growth ?

- (A) Dry and hot weather (B) Cold and dry weather
(C) Warm and humid weather (D) Windy conditions

Ans : (C) [FSO 2026]

Q.35 Pathogens that can survive without a host are called

- (A) Biotrophs (B) Necrotrophs
(C) Facultative saprophytes (D) Obligate parasites

Ans : (C) [FSO 2026]

Q.36 Bacterial leaf blight is a prevalent and destructive disease of one of the following agricultural crop

- (A) Rice (B)
(C) (D)

Ans : 2 — Rice [FSO 2020]

Q.37 _____ is popularly known as teak skeletonizer.

- (A) *Eutectona machaeralis* (B)
(C) (D)

Ans : 4 — *Eutectona machaeralis* [FSO 2020]

Q.38 _____ are root feeders which kill the seedlings by destruction of rootlets or removal of bark of tap root.

- (A) Cock chafers (B)
(C) (D)

Ans : 1 — Cock chafers [FSO 2020]

Q.39 Canker is a _____ symptom.

- (A) Necrotic (B)
(C) (D)

Ans : 3 — Necrotic [FSO 2020]

Q.40 Loranthus is a _____ which causes damage to forest trees.

- (A) Parasite (B)
(C) (D)

Ans : 1 — Parasite [FSO 2020]

Chemical, biological and genetic methods of disease control (including transgenic plants)

Q.41 RNA interference (RNAi) is used in transgenic plants to

- (A) Increase photosynthesis
(B) Silence specific pathogen genes
(C) Enhance nitrogen fixation
(D) Accelerate flowering

Ans : (B) [FSO 2026]

Plant Varieties: Origin, importance, export potential, varieties, climate, soil requirements, propagation and planting and after care

Q.42 Which one of the following is not a groundnut variety?

- (A) Chandramukhi (B)
(C) (D)

Ans : 2 — Chandramukhi [FSO 2020]

Q.43 What is the duration of protection of registered varieties of trees and vines?

- (A) 18 years (B)
(C) (D)

Ans : 2 — 18 years [FSO 2020]

Q.44 Which of the following have been provided to the farmers under the PPV & FR Act, 2001?

- (A) All of the given options (B)
(C) (D)

Ans : 4 — All of the given options [FSO 2020]

2. FORESTS & ECOLOGY

Definition of biodiversity

Q.1 The vegetation in the Rayalaseema region is primarily

- (A) Thorn scrub (B) Evergreen forest
(C) Grasslands (D) Wetlands

Ans : (A) [FSO 2026]

Q.2 The predominant forest type in Andhra Pradesh is

- (A) Tropical dry deciduous (B) Montane forests
(C) Mangrove (D)

Ans : (B) [FSO 2026]

Q.3 The animals dwelling in _____ ecosystem show gregariousness, keen sight and swift running.

- (A) Grassland (B)
(C) (D)

Ans : 1 — Grassland [FSO 2020]

Q.4 Animals which can tolerate narrow fluctuations in salt concentration are called as _____

- (A) Stenohaline (B)
(C) (D)

Ans : 2 — Stenohaline [FSO 2020]

Structure and functions of an ecosystem - role of Producers, consumers and decomposers

Q.5 What is the role of producers in an ecosystem?

- (A) Decompose dead material
(B) Hunt herbivores
(C) Create organic matter using sunlight
(D) Feed on other organisms

Ans : (C) [FSO 2026]

Q.6 What is the role of decomposers

- (A) Trap solar energy
(B) Eat other animals
(C) Break down dead matter and recycle nutrients
(D) Produce seeds

Ans : (C) [FSO 2026]

Q.7 _ In an ecosystem, if decomposers are removed, which of the following processes would be most directly affected first?

- (A) Photosynthesis (B) Nutrient cycling
(C) Food web stability (D) Predator-prey balance

Ans : (B) [FSO 2026]

Q.8 Which of the following best explains the functional role of producers in an ecosystem?
(i) They recycle minerals into the soil

- (A) They feed on dead organic matter
 (B) They regulate predator population
 (C) They convert solar energy into chemical energy
 (D)

Ans : (D) [FSO 2026]

Q.9 Keystone species are important because they

- (A) Always live in aquatic ecosystems
 (B) Maintain the diversity and stability of ecosystems
 (C) Have no ecological role
 (D) Haveno predators

Ans : (B) [FSO 2026]

Q.10 Animals which have a wide range of temperature tolerance are called

- (A) Eurythermal (B)
 (C) (D)

Ans : 3 — Eurythermal [FSO 2020]

Q.11 The role played by a species in its natural habitat is called

- (A) Niche (B)
 (C) (D)

Ans : 2 — Niche [FSO 2020]

Q.12are groups of species within biological communities having similar ecological features in a habitat.

- (A) Guilds (B)
 (C) (D)

Ans : 1 — Guilds [FSO 2020]

Q.13is the scientific study of animal behaviour under natural conditions.

- (A) Ethology (B)
 (C) (D)

Ans : 3 — Ethology [FSO 2020]

Q.14is a species that has a disproportionately large effect on its natural environment relative to its abundance.

- (A) Keystone species (B)
 (C) (D)

Ans : 2 — Keystone species [FSO 2020]

Ecological succession, Food chains, food webs and ecological pyramids

Q.15 There are _____ types of ecological pyramids.

- (A) 3 (B)
 (C) (D)

Ans : 4 — 3 [FSO 2020]

Q.16 Erosion, Physiography, Elevation and Subsidence are the _____ causes of Primary succession

- (A) Initial (B)
 (C) (D)

Ans : 1 — Initial [FSO 2020]

Biogeographical classification of India and value of biodiversity

Q.17 How many biogeographical zones are in India based on vegetation and geography?

- (A) 5climatezones
 (B) 10 zones based on vegetation and geography
 (C) 15 river systems
 (D) 12 zonesbased on vegetation and geography

Ans : (B) [FSO 2026]

Q.18 India's diverse climatic conditionsresult inwhich of thefollowing?

- (A) A uniform type of forest throughout the country
 (B) Limited vegetation and foresttypes
 (C) A wide range of forest types and plant species
 (D) Only evergreen forests

Ans : (C) [FSO 2026]

Q.19 Oriental realm for distribution of wildlife includes

- (A) Southern Arctic and Lower Africa
 (B) North America, Europe and Japan
 (C) Australia and New Zealand
 (D) India and Southern East Asia

Ans : (D) [FSO 2026]

Q.20 Cold seeps are areas on the ocean floor where

- (A) Hydrogen sulfide, methane and other hydrocarbon-rich fluids occur
 (B) Magma erupts
 (C) Photosynthesis takes place
 (D) Water evaporates

Ans : (A) [FSO 2026]

Q.21 According to Champion and Seth Classification, 1968, there are major groups of forests in India

- (A) Five (B)
 (C) (D)

Ans : 1 — Five [FSO 2020]

Q.22 India is divided into how many Bio-geographical zones?

- (A) 10 (B)
(C) (D)

Ans : 3 — 10 [FSO 2020]

Q.23 The tropical forests in India are differentiated into _____ groups

- (A) Seven (B)
(C) (D)

Ans : 2 — Seven [FSO 2020]

Biodiversity at global, National and local levels

Q.24 What is India's rank in the Global Nature Conservation Index 2024?

- (A) 1st (B) 176th
(C) 76th (D) 180th

Ans : (B) [FSO 2026]

Q.25 If climate change alters the rainfall pattern of a region, leading to replacement of moist deciduous forests by scrublands, what type of diversity is primarily impacted?

- (A) Genetic diversity (B) Species diversity
(C) Ecosystem diversity (D) Population diversity

Ans : (C) [FSO 2026]

Q.26 Which of the following methods is used for assessing genetic diversity?

- (A) All of the given options (B)
(C) (D)

Ans : 4 — All of the given options [FSO 2020]

Q.27 How many mega-biodiversity countries are there in the world?

- (A) 17 (B)
(C) (D)

Ans : 2 — 17 [FSO 2020]

Hot-spots of biodiversity

Q.28 Which is the first gazetted biosphere reserve in Andhra Pradesh, known for its endemic species and Eastern Ghats ecosystem ?

- (A) Coringa Wildlife Sanctuary
(B) Seshachalam Biosphere Reserve
(C) Pulicat Bird Sanctuary
(D) Papikonda National Park

Ans : (B) [FSO 2026]

Q.29 Naupada Coastal and Marine Biodiversity Area (ICMBA) is located in which district of Andhra Pradesh?

- (A) Vizianagaram (B) Visakhapatnam
(C) Srikakulam (D) Alluri Sitharama Raju

Ans : (C) [FSO 2026]

Q.30 A region is declared a biodiversity hotspot primarily based on which of the following criteria?

- (A) High forest cover and large area
(B) Presence of endemic species and high threat of habitat loss
(C) High rainfall and dense vegetation
(D) Presence of many exotic species

Ans : None [FSO 2026]

Q.31 Which of the following fresh water lake is also a bird sanctuary as well as a Ramsar wetland?

- (A) Kolleru lake, Andhra Pradesh
(B)
(C)
(D)

Ans : 2 — Kolleru lake, Andhra Pradesh [FSO 2020]

Threats to biodiversity such as habitat loss, poaching of wildlife, man-wildlife conflicts

Q.32 Which term describes a species that is not native to a specific area ?

- (A) Indigenous (B) Exotic
(C) Coppice (D) Rotation

Ans : (B) [FSO 2026]

Q.33 How many Tiger Reserves are there in India?

- (A) 50 (B)
(C) (D)

Ans : 1 — 50 [FSO 2020]

Q.34 Among the browsing animals, _____ is the most destructive animal destructing the forests.

- (A) Goat (B)
(C) (D)

Ans : 1 — Goat [FSO 2020]

Endangered and endemic species of India

Q.35 Which of the following is an endangered species in India?

- (A) House sparrow (B) Bengal tiger
(C) Cow (D) Domestic cat

Ans : (B) [FSO 2026]

Q.36 Which protected area is known for conserving a subspecies of the Indian swamp deer (Barasingha), adopted to hard ground and exclusively graminivorous?

- (A) Kanha National Park
(B) Manas National Park
(C) Mudumalai Wildlife Sanctuary
(D) TalChhapar Wildlife Sanctuary

Ans : (A) [FSO 2026]

Q.37 If a species found only in the Eastern Himalayas is classified as Vulnerable by the IUCN, and climate change reduces its habitat range, what could be its next likely status?

- (A) Least Concern (B) Near Threatened
(C) Endangered (D) Data Deficient

Ans : (C) [FSO 2026]

Q.38 Sri Lankamalleswara Wildlife Sanctuary is habitat for which of the critically endangered species ?

- (A) Great Indian Bustard (B) Jerdon's Courser
(C) Blue Jay (D) Peafowl

Ans : (B) [FSO 2026]

Q.39 Which wild animal is typical of India's cold desert region?

- (A) Lion (B) Snow leopard
(C) Tiger (D) Bear

Ans : (B) [FSO 2026]

Q.40 How many sub-species of Tigers have been now recognised based on Molecular markers?

- (A) 6 (B)
(C) (D)

Ans : 2 — 6 [FSO 2020]

Q.41 As per IUCN Red list Snow leopard is categorised under _____ list.

- (A) Vulnerable (B)
(C) (D)

Ans : 3 — Vulnerable [FSO 2020]

Q.42 _____ is an extinct species of India.

- (A) Asiatic Cheetah (B)
(C) (D)

Ans : 4 — Asiatic Cheetah [FSO 2020]

In-situ and Ex-situ conservation of biodiversity

Q.43 _____ are constituted on government lands which are adjacent to national parks and sanctuaries especially linking one protected area with another

- (A) Conservation reserve (B)
(C) (D)

Ans : 3 — Conservation reserve [FSO 2020]

Q.44 The following is a site of ex-situ conservation

- (A) Botanical garden (B)
(C) (D)

Ans : 3 — Botanical garden [FSO 2020]

Convention of Biological Diversity (CBD)

Q.45 In which year was the Biological Diversity Act enacted in India ?

- (A) 2000 (B) 2002
(C) 2010 (D) 2017

Ans : (B) [FSO 2026]

Role of afforestation and forest regeneration in carbon sequestration

Q.46 How does afforestation help in climate change mitigation ?

- (A) Produces more wood
(B) Destroys pests
(C) Absorbs CO through photosynthesis
(D) Causes flooding

Ans : (C) [FSO 2026]

Q.47 Carbon stock assessment is part of

- (A) Legal forest protection (B) Climate impact evaluation
(C) Timber pricing (D) Forest road design

Ans : (B) [FSO 2026]

Q.48 Forestry aimed at producing flow of protection and recreation benefits to the community is

- (A) Agro forestry (B) Urban forestry
(C) Social forestry (D) Environmental forestry

Ans : (C) [FSO 2026]

3. HORTICULTURE

Definition and different branches of horticulture

Q.1 What is the science of vegetable production called?

- (A) Olericulture (B)
(C) (D)

Ans : 3 — Olericulture [FSO 2020]

Q.2 What are EPP-1, EPP-2, EPP-3, EPP-4 and EPP-5?

- (A) Varieties of (B)
Cashewnut
(C) (D)

Ans : 4 — Varieties of Cashewnut [FSO 2020]

Q.3 What is the best suited soil type for cultivation of Cashewnut?

- (A) Red sandy loam (B)
(C) (D)

Ans : 2 — Red sandy loam [FSO 2020]

Q.4 Fruit canker is a post harvest disease in which of the following crops?

- (A) Guava (B)
(C) (D)

Ans : 4 — Guava [FSO 2020]

Importance of horticulture in terms of economy, production, employment generation, environmental protection and human resource development

Q.5 Which sector shows the strongest link between horticultural development and foreign exchange earnings?

- (A) Export of cereals (B) 3. Export of pulses
(C) (D) Export of livestock

Ans : (B) [FSO 2026]

Q.6 Where is ICAR - Central Institute of Subtropical Horticulture located?

- (A) Lucknow (B)
(C) (D)

Ans : 4 — Lucknow [FSO 2020]

Q.7 Which of the following portals for farmers enables all Central and State government organisations in agriculture and allied sectors to give information / services / advisories to farmers by SMS in their language regarding preference of agricultural practices and location?

- (A) mKisan Portal (B)
(C) (D)

Ans : 1 — mKisan Portal [FSO 2020]

Q.8 In which of the following plantation crops Eriophyid is a major pest?

- (A) Coconut (B)
(C) (D)

Ans : 2 — Coconut [FSO 2020]

Nutritional value of horticultural crops, Divisions of horticulture and their importance

Q.9 Which branch of horticulture deals with fruit crops ?

- (A) Olericulture (B) Pomology
(C) Floriculture (D) Arboriculture

Ans : (B) [FSO 2026]

Q.10 What is the use of papain, a proteolytic enzyme, present in the latex obtained from immature fruits of papaya?

- (A) For meat (B)
tenderization
(C) (D)

Ans : 3 — For meat tenderization [FSO 2020]

Propagation of plants by cuttage, Types of cuttings, and the factors affecting regeneration of plants from cuttings

Q.11 The success of root regeneration in cuttings is most strongly influenced by which (S) physiological factor ? (I) High carbohydrate reserves and hormonal balance

- (A) Seed viability
(B) Leaf venation pattern
(C) Flowering stage of the mother plant
(D)

Ans : (B) [FSO 2026]

Propagation by layerage and the factors affecting regeneration of plants by layerage and the most common methods of layerage

Q.12 Which plant is commonly propagated by mound layering ?

- (A) Rose (B) Apple rootstocks
(C) Banana (D) Pineapple

Ans : (B) [FSO 2026]

Q.13 Propagation through tissue culture belongs to

- (A) Sexual propagation (B) Macropropagation
(C) Micropropagation (D) Hybrid propagation

Ans : (C) [FSO 2026]

Q.14 Which environmental factor most strongly influences successful rooting in layerage?
(I) Very low soil moisture

- (A) High humidity around the rooting zone
(B) High wind speed
(C) Short day length only
(D)

Ans : (B) [FSO 2026]

Propagation by grafting, importance of graftage and the factors for successful grafting, Selection of rootstock and scion, methods of budding and grafting

Q.15 Which is a common rootstock for apple grafting ?

- (A) M9 dwarfing (B) Juglans regia
(C) Oryza sativa (D) Zea mays

Ans : (A) [FSO 2026]

Q.16 A critical requirement for successful grafting is
(I) Alignment of cortical tissues

- (A) Proper alignment of cambial layers of scion and stock
(B) Removal of all buds from scion
(C) Use of genetically unrelated species
(D)

Ans : (B) [FSO 2026]

Q.17 Rootstock selection is important because it influences

- (A) Only fruit color
(B) Tree vigor, stress tolerance, disease resistance and yield
(C) Only flower size
(D) Only plant height without other effects

Ans : (B) [FSO 2026]

Q.18 Which of the following horticultural crop is propagated by Stone grafting method?

- (A) Mango (B)
(C) (D)

Ans : 2 — Mango [FSO 2020]

Importance and scope, production of crops in greenhouse

Q.19 Hydroponics is often integrated with greenhouses to grow plants

- (A) In soil-less media (B) Under shade nets
(C) In open fields (D) Without water

Ans : (A) [FSO 2026]

Q.20 Orientation of single greenhouses with latitude above 40°N should have ridge running

- (A) East to West (B)
(C) (D)

Ans : 2 — East to West [FSO 2020]

Q.21 is a general term which applies to the growing of plants without soil in an inert medium into which nutrient solutions are usually pumped automatically at regular intervals.

- (A) Gravel culture (B)
(C) (D)

Ans : 3 — Gravel culture [FSO 2020]

Q.22 Hygrometer in a greenhouse is used for measuring

- (A) Relative humidity (B)
(C) (D)

Ans : 4 — Relative humidity [FSO 2020]

4. SOIL SCIENCE

Basics of earth, rocks and minerals

Q.1 Which rocks are formed due to the deposition and compaction of the remains of plants ?

- (A) Calcareous rocks (B) Carbonaceous rocks
(C) Siliceous rocks (D) Siltstone rocks

Ans : (B) [FSO 2026]

Q.2 Which of the following is an example of an igneous rock?

- (A) Lime stone (B) Basalt.
(C) Shale (D) Gypsum /)

Ans : (B) [FSO 2026]

Study of folds, joints, faults, foliation and lineation in geological formation

Q.3 Rift valleys are formed due to

- (A) Faults (B) Folds
(C) Joints (D) Foliations

Ans : (A) [FSO 2026]

Q.4 A break or fracture in the Earth's Crust is called

- (A) Fold (B) Joint
(C) Fault (D) Lineation

Ans : (C) [FSO 2026]

Q.5 The dip of the bedding plane measured in a direction other than that of true dip is known as

- (A) Apparent dip (B)
(C) (D)

Ans : 2 — Apparent dip [FSO 2020]

Q.6 A line drawn on a bedding plane at right angles to the direction of true dip is known as

- (A) Strike line (B)
(C) (D)

Ans : 1 — Strike line [FSO 2020]

Q.7 Exposed surface of a rock stratum on the ground is known as an

- (A) Outcrop (B)
(C) (D)

Ans : 1 — Outcrop [FSO 2020]

Types of soils, their field identification and classification

Q.8 What is the another name of 'Black Soils'?

- (A) Podzol (B) Latosol
(C) Chernozem (D) Azonal

Ans : (C) [FSO 2026]

Q.9 Which soil grown in humid areas under rich vegetation cover ?

- (A) Pedocols (B) Chestnutsoil
(C) Latosol (D) Pedalfars

Ans : (D) [FSO 2026]

Forest soils and their classification, factors affecting soil formation

Q.10 Which of the following is not considered a topographic factor?

- (A) Rainfall (B) Slope
(C) Aspect (D) Altitude

Ans : (A) [FSO 2026]

Physical, chemical and biological properties of soils

Q.11 The study of regolith is important in forestry because

- (A) It indicates rainfall patterns
(B) It reduces water infiltration
(C) It prevents afforestation
(D) It determines soil fertility and root penetration

Ans : (D) [FSO 2026]

Q.12 The primary functions of geotextiles in soil are

- (A) Separation, filtration, reinforcement, protection, drainage
(B) Decoration and aesthetics
(C) Fertilization and irrigation
(D) Weather forecasting

Ans : (A) [FSO 2026]

Q.13 Which of the following conditions define 'Saline soil'?

- (A) $EC_e > 4 \text{ dS/m}$, $pH < 8.5$ and $ESP < 15$
(B)
(C)
(D)

Ans : 4 — $EC_e > 4 \text{ dS/m}$, $pH < 8.5$ and $ESP < 15$ [FSO 2020]

Q.14is the size of clay particles as per USDA classification of soil texture.

- (A) $< 0.002\text{mm}$ (B)
(C) (D)

Ans : 4 — $< 0.002\text{mm}$ [FSO 2020]

Q.15 Which fraction of the following water is readily available to plants?

- (A) Capillary water (B)
(C) (D)

Ans : 3 — Capillary water [FSO 2020]

Q.16evaluates the quality of a soil as a function of its characteristics, water, plant and other biological properties.

- (A) Soil Health Card (B)
(C) (D)

Ans : 3 — Soil Health Card [FSO 2020]

Q.17 _____ is the mechanical movement of clay and other fine particles down the profile.

- (A) Eluviation (B)
(C) (D)

Ans : 4 — Eluviation [FSO 2020]

Soil conservation — definition and causes for erosion, types of erosion, agents of erosion

Q.18 Excessive use of chemical fertilizers leads to

- (A) Soil fertility improvement aa
- (B) Increased biodiversity
- (C) Soil degradation
- (D) Alkalization only

Ans : (C) [FSO 2026]

Q.19 When is the World Soil Day celebrated annually?

- (A) 05th December (B)
- (C) (D)

Ans : 3 — 05th December [FSO 2020]

Q.20 Which of the following is Primary tillage?

- (A) All of the given options (B)
- (C) (D)

Ans : 4 — All of the given options [FSO 2020]

Q.21 Which of the following is the most severe form of the erosion?

- (A) Gully erosion (B)
- (C) (D)

Ans : 2 — Gully erosion [FSO 2020]

Reclamation of saline and alkaline soils, water logged and other waste lands

Q.22 Which of the following are reclamation methods of alkaline and saline soils ? cofom (I) Improved irrigation and soil drainage

- (A) Decreasing water and wet
- (B) Cultivation of crops and ploughing
- (C) Keeping under fallow conditions and storing rain water n
- (D)

Ans : (B) [FSO 2026]

Q.23refers to a technique of lowering groundwater in waterlogged areas by raising tree plantations.

- (A) Bio-drainage (B)
- (C) (D)

Ans : 2 — Bio-drainage [FSO 2020]

Q.24 Lime, dolomite, basic slag, flue dust and wood ash are used for reclamation of

- (A) Acidic soil (B)
- (C) (D)

Ans : 2 — Acidic soil [FSO 2020]

Role of forests in conserving soils

Q.25 In which areas afforestation is important soil conservation method ?

- (A) Glacial areas
- (B) Semi arid and low rainfall receiving areas
- (C) Very high rainfall receiving areas
- (D) Cropped areas

Ans : (B) [FSO 2026]

5. WATER RESOURCE MANAGEMENT

Surface and subsurface water resources

Q.1 What is the live storage capacity (approx.) of the Pulichintala Reservoir ?

- (A) 20 Tmcft (B) 36 Tmcft
- (C) 46 Tmcft (D) 80 Tmcft

Ans : (C) [FSO 2026]

Impurities of water and their significance

Q.2 The hardness of water is primarily due to the presence of which ions?

- (A) Sodium and chloride (B) Calcium andmagnesium
- (C) Strontium and potassium) (D) Potassium and nitrate

Ans : (B) [FSO 2026]

Q.3 Water with less than 1000 mg/lof Total Dissolved Solids (TDS) is classified as

- (A) Brine water (B) Fresh water
- (C) Slightly saline (D) Moderately saline

Ans : (B) [FSO 2026]

Q.4 Which type of impurity in water is biologically significant because it directly influences oxygen demand and microbial growth?

- (A) Carbonate minerals (B) Dissolved organic matter
- (C) Ironandmanganese (D) Suspended silica

Ans : (B) [FSO 2026]

Q.5 Forest water contains the following chemical in highest concentration

- (A) MgO (B) CaO
- (C) K2O (D) Fe,0

Ans : (B) [FSO 2026]

Q.6 Blue baby syndrome is high concentration ofin drinking water.

- (A) Nitrate (B)
(C) (D)

Ans : 1 — Nitrate

[FSO 2020]

Physical, chemical and bacteriological analysis, water borne diseases

Q.7 Under which Section of the Water (Prevention and Control of Pollution) Act, 1974 can a State Pollution Control Board serve directions to industries or persons for preventing or controlling water pollution ?

- (A) Section 30 (B) Section 20
(C) Section 33 (D) Section 21

Ans : (B)

[FSO 2026]

Q.8 Which of the following water-borne diseases is caused by a protozoan parasite rather than bacteria or virus?

- (A) Cholera (B) Hepatitis A
(C) Amoebic dysentery (D) Typhoid

Ans : (C)

[FSO 2026]

Q.9 Which physical parameter of water most directly indicates suspended impurities and affects light penetration in lakes and reservoirs ?

- (A) Colour (B) Turbidity
(C) Odour (D) Temperature

Ans : (B)

[FSO 2026]

Q.10 In Physical impurity tests, water colour is expressed inunits.

- (A) Hazen units (B)
(C) (D)

Ans : 2 — Hazen units

[FSO 2020]

Pumping and gravity schemes

Q.11 Which law mandates that every state shall constitute a Water Resources Regulatory Authority?

- (A) Environmental Protection Act, 1986
(B) National Water Policy, 2012
(C) State-specific Water Resources Acts
(D) Biological Diversity Act, 2002

Ans : (C)

[FSO 2026]

Q.12 The Andhra Pradesh, Water Land and Trees Act (WALTA) was enacted in which year ?

- (A) 1998 (B) 2002
(C) 2005 (D) 2010

Ans : (B)

[FSO 2026]

6. WATERSHED MANAGEMENT

Role of mini-forests and forest trees in overall resource management

Q.1 Which tree species is commonly recommended in watershed afforestation due to its fast growth and soil-binding roots ?

- (A) *Acacia nilotica* (B) *Eucalyptus globulus*
(C) *Pinus roxburghii* (D) *Tectona grandis*

Ans : (A)

[FSO 2026]

Water-harvesting and Soil conservation

Q.2 Checkdams, contour bunding and percolation tanks are associated with

- (A) Air pollution control
(B) Groundwater depletion
(C) Watershed development and water harvesting
(D) Industrial effluent treatment

Ans : (C)

[FSO 2026]

Q.3 The Integrated Watershed Management Programme (IWMP) is implemented under which Ministry?

- (A) Ministry of Jal Shakti
(B) Ministry of Rural Development
(C) Ministry of Environment, Forest and Climate Change
(D) Ministry of Agriculture

Ans : (B)

[FSO 2026]

Q.4 Which of the following best explains how forests reduce flood intensity in watershed areas?

- (A) They absorb rainfall and slow down runoff through dense vegetation and root networks
(B) They divert river channels artificially
(C) They store all rainwater permanently in the canopy
(D) They increase surface erosion, reducing storage capacity

Ans : None

[FSO 2026]

7. GENERAL SILVICULTURE

Definition and principles of General Silviculture

Q.1 The branch of forestry that deals with theory and practice of raising forest crops is

- (A) Forest utilization (B) Agroforestry
(C) Silviculture (D) Forest protection

Ans : (C)

[FSO 2026]

Q.2 their growth and after care up to the time of final harvesting is referred as

- (A) Forest Mensuration (B) Forest Management
(C) Forest Utilization (D) Silviculture

Ans : (D) [FSO 2026]

Q.3 Which of the following correctly describes the uses of Santalum album?

- (A) Mainly valued for its latex and gum used in adhesives
(B) Known for fragrant heartwood and oil used in perfumery, cosmetics, medicines and
(C) Cultivated primarily as a fodder tree for livestock
(D) Used only as a fuelwood species with no commercial significance

Ans : None [FSO 2026]

Q.4 Which branch of forestry deals with the establishment, development, care and reproduction of forest crops especially timber ?

- (A) Dendrology (B) Agroforestry
(C) Silviculture (D) Forestmensuration

Ans : (C) [FSO 2026]

Q.5 Which of the following species is most suitable for shelterbelt plantation in coastal areas?

- (A) Mango (B) Teak
(C) Rosewood (D) Casuarina

Ans : (D) [FSO 2026]

Q.6 Red Sanders is endemic to which region of India and has high demand in the international market?

- (A) Western Ghats
(B) Southern Part of Eastern Ghats
(C) Himalayas
(D) Aravalli Hills

Ans : (B) [FSO 2026]

Q.7 Arabica gum is extracted from which species and is mainly used for?

- (A) Azadirachta indica (B) Santalum album
(C) Tectona grandis (D) Acacia nilotica

Ans : (D) [FSO 2026]

Q.8 The unusable residue lying on the forest floor after logging is called

- (A) Slash (B)
(C) (D)

Ans : 2 — Slash [FSO 2020]

Ecological and physiological factors influencing vegetation, natural and artificial regeneration of forests

Q.9 In the natural regeneration process seeds of red sanders is normally dispersed through

- (A) Water (B) Wind
(C) Birds (D) Gravity

Ans : (B) [FSO 2026]

Q.10 Cutting back of grown up trees to the ground level is known as

- (A) Tending (B) Singling
(C) Coppicing (D) Brashing

Ans : (C) [FSO 2026]

Q.11 is a species of inferior quality or size, of relatively little Silvicultural value or importance, associated with the principal species.

- (A) ACCESSORY SPECIES (B)
(C) (D)

Ans : 1 — ACCESSORY SPECIES [FSO 2020]

Q.12 is a method of thinning carried out in the irregular crop.

- (A) Selection thinning (B)
(C) (D)

Ans : 4 — Selection thinning [FSO 2020]

Q.13 Which one of the followings is not a tending operation?

- (A) Regeneration felling (B)
(C) (D)

Ans : 1 — Regeneration felling [FSO 2020]

Q.14 The rapid diameter increment of trees due to light stimulus following increment fellings is known as

- (A) Light increment (B)
(C) (D)

Ans : 2 — Light increment [FSO 2020]

Q.15 Nitrogen fixing ability of Casuarina is attributable to

- (A) Frankia (B)
(C) (D)

Ans : 2 — Frankia [FSO 2020]

Q.16 Controlled burning in a forests is

- (A) Silvicultural operation (B)
(C) (D)

Ans : 3 — Silvicultural operation [FSO 2020]

Q.17 Forest fire spreading slowly over the ground with low flame is called _____ fire.

- (A) Creeping (B)
(C) (D)

Ans : 4 — Creeping [FSO 2020]

Q.18 Water, Earth, Beating and Counter-firing are the methods of _____ for control of fire.

- (A) extinguishing (B)
measures
(C) (D)

Ans : 2 — extinguishing measures [FSO 2020]

Economic importance of some of the forestry tree species of India: *Acacia Sundra, Acacia nilotica, Albizzia lebbeck, Albizzia procera, Anthocephalus Cadamba, Anogeissus latifolia, Azadirachta indica, Bamboo spp, Butea monosperma, Casuarina equisetifolia, Dalbergia sisoo, Emblica officinalis, Eucalyptus spp, Gmelina arborea, Hardwickia binata, Lagerstremia lanceolata, Pterocarpus marsupium, Pterocarpus santalinis, Santalum album, Tectona grandis, Terminalia tomentosa, Tamarindus indica*

Q.19 Which forest product is used in pharmaceutical and cosmetic industries ?

- (A) Sal wood (B) Gum and resins
(C) Sandalwood logs (D) Mahogany timber

Ans : (B) [FSO 2026]

Q.20 The following tree is propagated through rhizomes

- (A) Bamboo (B) Teak
(C) Eucalyptus (D) Terminalia

Ans : (A) [FSO 2026]

Q.21 Tree species successfully growth on large scale in coastal sandy soils are

- (A) Casuarina (B) Acacia
(C) Eucalyptus (D) Azadirachta

Ans : (A) [FSO 2026]

Q.22 Which industry consumes a major proportion of bamboo ?

- (A) Dye and tanning industry
(B) Paper manufacturing industry
(C) Oil and bio diesel industry
(D) Traditional medicine

Ans : (B) [FSO 2026]

Q.23 Kath (Catechu) is prepared from which tree species?

- (A) *Acacia nilotica* (B) *Acacia chundra*
(C) *Azadirachta indica* (D) *Ficus religiosa*

Ans : (B) [FSO 2026]

8. AGRO-FORESTRY

Agro forestry, its definition, scope and necessity

Q.1 is the ability of a grassland unit to give adequate support to a constant number of livestock for a stated period each year without deteriorating with respect to this and / or other proper land use.

- (A) Carrying capacity (B)
(C) (D)

Ans : 2 — Carrying capacity [FSO 2020]

Q.2 The sustainable land use systems involving trees combined with crops and / or animals on the same unit of land is termed as

- (A) Agroforestry (B)
(C) (D)

Ans : 1 — Agroforestry [FSO 2020]

Agro forestry systems under different agro-ecological zones, selection of species and role of multipurpose trees and NTFPs, techniques, food, fodder and fuel security

Q.3 Which type of farming is used to restore soil fertility in India's dry lands?

- (A) Ley farming (B)
(C) (D)

Ans : 3 — Ley farming [FSO 2020]

Q.4 Home Gardens is the most prevalent agro-forestry system being adopted in

- (A) Humid tropical region (B)
(C) (D)

Ans : 1 — Humid tropical region [FSO 2020]

Q.5 _____ species is an important fodder tree for dry region of the country.

- (A) *Acacia nilotica* (B)
(C) (D)

Ans : 4 — *Acacia nilotica* [FSO 2020]

Q.6 _____ is called as 'alfalfa of the tropic' because of its nutritious fodder.

- (A) *Leucaena leucocephala* (B)
(C) (D)

Ans : 1 — *Leucaena leucocephala* [FSO 2020]

Q.7 Among the crops grown in the homestead _____ is the most dominant and important tree known as 'Tree of Hundred uses'

- (A) Coconut (B)
(C) (D)

Ans : 3 — Coconut [FSO 2020]

Q.8 The _____ homegardens are cited as an excellent example to show diversity and complexity of the structure and function of tropical homegardens.

- (A) Javanese (B)
(C) (D)

Ans : 1 — Javanese [FSO 2020]

Q.9 _____ is described as "wood of the poor" in India, "friend of the people" in China and "brother" in Vietnam

- (A) Bamboo (B)
(C) (D)

Ans : 1 — Bamboo [FSO 2020]

9. SOCIAL FORESTRY AND COMMUNITY PARTICIPATION

Social Forestry: The objectives, scope and necessity of social forestry including the urban forestry in Andhra Pradesh

Q.1 The word Social Forestry was first pronounced by

- (A) Swaminathan (B) Jadhav Payeng
(C) Westoby (D) Brandis

Ans : (C) [FSO 2026]

Q.2 Which of the following is not a type of social forestry?

- (A) Farm forestry (B) Community forestry
(C) Urban forestry (D) Industrial forestry

Ans : (D) [FSO 2026]

Aim and objectives, principles, methodology and benefits of AP community forest management

Q.3 Which forest protection strategy helps in community participation for conserving local forests?

- (A) Joint Forest Management (B) Monoculture Plantations
(C) Mining in Forest Regions (D) Wildlife Tourism Expansion

Ans : (A) [FSO 2026]

Q.4 The village level forest protection committees also called as

- (A) Wildlife Conservation Committees
(B) Water Conservation Committees
(C) Land Conservation Committees
(D) Vana Samrakshana Samithies

Ans : (D) [FSO 2026]

Q.5 The first JFM guidelines was issued by MoEF in the year

- (A) 1990 (B)
(C) (D)

Ans : 1 — 1990 [FSO 2020]

10. ANIMAL HEALTH & NUTRITION

Role of nutrition in animal health and production

Q.1 Which nutrient provides the highest energy content to animals ?

- (A) Carbohydrates (B) Proteins
(C) Fats (D) Fiber

Ans : (C) [FSO 2026]

Q.2 Which animal-source food has the highest protein content?

- (A) Egg (B) Milk
(C) Meat (D) Fish

Ans : (C) [FSO 2026]

Q.3 Panchali is a breed of

- (A) Sheep (B)
(C) (D)

Ans : 2 — Sheep [FSO 2020]

Q.4 Actual milk yield is converted into milk with 4% fat by using formula

Where, FCM = Fat corrected milk; M = weight of milk; F = weight of fat in milk

- (A) $FCM = 0.4M + 15F$ (B)
(C) (D)

Ans : 1 — $FCM = 0.4M + 15F$ [FSO 2020]

Q.5 _____ is an Indian sub-continent Milch Breed.

- (A) Sahiwal (B)
(C) (D)

Ans : 4 — Sahiwal [FSO 2020]

Q.6 _____ breed among the Indian Goats is from North-Western Arid and Semi-Arid region.

- (A) Sirohi (B)
(C) (D)

Ans : 3 — Sirohi [FSO 2020]

Conservation of feeds and fodder and utilization of agro by-products

Q.7 Urea contains _____ % of nitrogen

- (A) 46 (B)
(C) (D)

Ans : 2 — 46 [FSO 2020]

Q.8 Of succulent feeds, _____ is the most convenient and economical method for maintaining larger livestock.

- (A) Pasture (B)
(C) (D)

Ans : 2 — Pasture [FSO 2020]

Feed supplements and additives

Q.9 Which of the following is a feed supplement ?

- (A) Terramycin (B) Vitablend
(C) Zinc bacitracin (D) Flavomycin

Ans : (B) [FSO 2026]

Q.10 _____ is an micro-ingredient included in animal feed in relatively small amounts with specific physiological and nutritional value.

- (A) Pronutrient (B)
(C) (D)

Ans : 3 — Pronutrient [FSO 2020]

Q.11 _____ is an example for plant protein as a source of animal feed.

- (A) Cotton seed cake (B)
(C) (D)

Ans : 1 — Cotton seed cake [FSO 2020]

Nutrition deficiencies and their management

Q.12 Deficiency of which Vitamin causes anaemia?

- (A) Vitamin E (B) Vitamin D
(C) VitaminB3 (D) VitaminB12

Ans : (D) [FSO 2026]

Q.13 Rickets is caused by the deficiency of

- (A) Calcium (B)
(C) (D)

Ans : 1 — Calcium [FSO 2020]

Major contagious diseases affecting cattle, buffaloes, sheep and goats

Q.14 Biological vector of bluetongue virus

- (A) Mosquito (B) Ticks
(C) Flea (D) Culicoides

Ans : (D) [FSO 2026]

Q.15 Which test is used to identify the mastitis disease in cattle ?

- (A) Widal test (B) Tetrazolium test
(C) Strep cup test (D) None of the above

Ans : (C) [FSO 2026]

Q.16 Which contagious disease causes chronic infection of the small intestine in cattle, sheep and goats?

- (A) Johne's disease (B) Anthrax
(C) Blackquarter (D) Rabies

Ans : (A) [FSO 2026]

Q.17 Tuberculosis in wild animals is primarily caused by

- (A) Mycobacterium bovis (B) Mycoplasma gallisepticum
(C) Salmonella typhi (D) Bacillus anthracis

Ans : (A) [FSO 2026]

Q.18 A disease of animals that can be transmitted to humans is called a

- (A) Zoonosis (B)
(C) (D)

Ans : 3 — Zoonosis [FSO 2020]

Q.19 _____ is an infectious disease of birds caused by influenza A virus.

- (A) Avian influenza (B)
(C) (D)

Ans : 1 — Avian influenza [FSO 2020]

Animal Feed management in Zoological parks

Q.20 The Nandi in Siva temples is said to resemble which breed of cattle?

- (A) Ongole (B)
(C) (D)

Ans : 2 — Ongole [FSO 2020]

11. ECONOMIC ZOOLOGY

Beneficial and harmful insects including insect vectors of human diseases, Industrial fish, prawn and molluscs of India, Non-poisonous and poisonous snakes of India, Venomous animals-centipede, wasp, honey bee

Q.1 How can you identify a python based on the nature of its ventrals?

- (A) Broad ventrals not extending across the belly
(B) Broad ventrals extending across the belly
(C) Small and narrow ventrals
(D) Presence of loreal pit

Ans : (A) [FSO 2026]

Q.2 Which insect acts as an important biological control agent by preying on pest insects ?

- (A) Braconid wasps (B) Termites
(C) Leaf hoppers (D) White flies

Ans : (A) [FSO 2026]

Q.3 The mission of which of the following organisations is "to ensure that the trade in wild plants and animals is not a threat to the conservation of nature"?

- (A) TRAFFIC (B)
(C) (D)

Ans : 2 — TRAFFIC [FSO 2020]

Q.4 The animals under this class have moist, soft skin and lack scales

- (A) Amphibia (B)
(C) (D)

Ans : 4 — Amphibia [FSO 2020]

Q.5 Phylum _____ includes animals like starfish, Sea urchin and Sea Star.

- (A) Echinodermata (B)
(C) (D)

Ans : 2 — Echinodermata [FSO 2020]

Q.6 _____ are viviparous snakes.

- (A) Viper (B)
(C) (D)

Ans : 2 — Viper [FSO 2020]

Q.7 _____ is a venomous snake.

- (A) Green Pit-Viper (B)
(C) (D)

Ans : 4 — Green Pit-Viper [FSO 2020]

Basics on the diseases caused by aberrant chromosomes/genes in man

Q.8 A well-known autosomal abnormality at birth is

- (A) Down's syndrome (B) Patau's syndrome
(C) Turner's syndrome (D)

Ans : (B) [FSO 2026]

DNA as a tool for forensic investigation

Q.9 The first technique that was adopted for forensic DNA analysis was

- (A) RFLP (B) PCR
(C) HLADQAI (D) STR

Ans : (A) [FSO 2026]

Q.10 What type of DNA commonly used in forensic identification?

- (A) Nuclear DNA
(B) Mitochondrial DNA
(C) Both Nuclear DNA and Mitochondrial DNA
(D) Only RNA

Ans : (C) [FSO 2026]

12. FOREST ECONOMICS

Basic fundamental principles of forest economics

Q.1 Who are the important stakeholder in the forest enterprise and management?

- (A) 2. Vana Samrakshana Samithies
(B)
(C) Cultural Values Promoters
(D) Forest Labour

Ans : (A) [FSO 2026]

13. FOREST LEGISLATION & ADMINISTRATION

Basic objectives and essence of important Forest Laws: AP Forest Act-1967, Wildlife Protection Act-1972, Forest Conservation Act-1980, Forest Rights Act-2006

Q.1 According to the National Forest Policy forest area in comparison to the total s't v geographical area should be

- (A) 1/3rd (B) 1/4th
(C) 1/6th (D)

Ans : (A) [FSO 2026]

Q.2 As per the A.P. Forest Act, 1967 (Amendment), a private persons possessing Red Sanders Wood above what quantity (in kgs) must inform their stocks to the concerned divisional Forest Officer within 180 days?

- (A) 5 kgs (B) 10 kgs
(C) 20 kgs (D) 25 kgs

Ans : (C) [FSO 2026]

Q.3 Which Authority is primarily responsible for initiating the process of recognizing forest rights under the RoFR Act, 2006?

- (A) Gram Sabha
(B) District Collector
(C) State Government
(D) Ministry of Environment, Forest and Climate Change

Ans : (A) [FSO 2026]

Q.4 Which section of Wildlife Protection Act, 1972 deals with declaration of National Parks?

- (A) Section 35 (B)
(C) (D)

Ans : 2 — Section 35 [FSO 2020]

Q.5 In which year, Andhra Pradesh Forest Act came into force?

- (A) 1967 (B)
(C) (D)

Ans : 2 — 1967 [FSO 2020]

Q.6 The Concurrent List includes which of the following?

- (A) All of the given options (B)
(C) (D)

Ans : 4 — All of the given options [FSO 2020]

Q.7 Which year the Forest Rights Act was enacted?

- (A) 2006 (B)
(C) (D)

Ans : 2 — 2006 [FSO 2020]

Q.8 _____ is not included in the definition of forest produce under section (2) of the Indian Forest Act, 1927

- (A) Fish (B)
(C) (D)

Ans : 1 — Fish [FSO 2020]

Q.9 The Wild Birds and Animals (Protection) Act was enacted in the year _____

- (A) 1912 (B)
(C) (D)

Ans : 3 — 1912 [FSO 2020]

Q.10 An area is declared as _____ under 26 A of the Wildlife (Protection) Act, 1972.

- (A) Sanctuary (B)
(C) (D)

Ans : 1 — Sanctuary [FSO 2020]

Q.11 _____ is one of the organisation involved in investigating wildlife crime in India

- (A) None of the given options (B)
(C) (D)

Ans : 4 — None of the given options [FSO 2020]

The structure and activities of the AP state Forest department: The genesis and objectives of Indian Forest Services, AP State Forest Services, AP State Forest Subordinate Services

Q.12 Which training institution is responsible for imparting foundational training to IFS probationers?

- (A) Forest Research Institute
(B) Indian Institute of Forest Management
(C) Indira Gandhi National Forest Academy
(D) Wildlife Institute of India

Ans : None [FSO 2026]

Q.13 The administrative head of Andhra Pradesh Forest Department is

- (A) Principal Chief Conservator of Forests & HoFF
(B) Additional Principal Chief Conservator of Forests
(C) Chief Conservator of Forests
(D) Conservator of Forests

Ans : (A) [FSO 2026]

14. FOREST MENSURATION

Methods of measuring diameter, girth, height and volume of trees

Q.1 Which of the following instrument is used to measure angle of slope normally used in determination of height of trees?

- (A) Hypsometer (B) Altimeter
(C) Clinometer (D) Girding tape

Ans : (C) [FSO 2026]

Q.2 Which instrument is commonly used to measure the Diameter at Breast Height (DBH) of a tree ?

- (A) Abney level (B) Haga Altimeter
(C) Calipers (D) Increment Borer

Ans : (C) [FSO 2026]

Q.3 In Forest Mensuration Breast Height (BH) in India is taken as

- (A) 1.30m (B) 1.50m
(C) 1.20 m (D) 1.37 m

Ans : (D) [FSO 2026]

Q.4 Which of the following instruments is not commonly used for measuring the diameter or girth of a tree?

- (A) Bark Gauge (B) Wooden Scale
(C) Callipers (D) Measuring Tape

Ans : (A) [FSO 2026]

Q.5 _____ maps are generally prepared on 1:50000 scale.

- (A) Management (B)
(C) (D)

Ans : 1 — Management [FSO 2020]

Q.6 _____ is the instrument used to measure the height of trees based on Trigonometric principles

- (A) Brandis Hypsometer (B)
(C) (D)

Ans : 1 — Brandis Hypsometer [FSO 2020]

Q.7 Girth Over Bark (GOB) can be converted into Girth Under Bark (GUB) by using the formula

- (A) $g = g' - 2 \pi t$ (B)
(C) (D)

Ans : 1 — $g = g' - 2 \pi t$ [FSO 2020]

Q.8 Diopter, basal area factor, plot radius factor, tree factor, calibration distance factor are terms used in

- (A) Point sampling (B)
(C) (D)

Ans : 4 — Point sampling [FSO 2020]

Form-factor

Q.9 'Form Factor' of trees is useful for

- (A) To estimate volume of standing trees
(B) To measure the height of trees
(C) To estimate volume of felled trees
(D) To measure the diameter of trees

Ans : (A) [FSO 2026]

Current annual increment versus mean annual increment

Q.10 Stump analysis studies the progress of _____ increment on stump throughout the life of the tree.

- (A) Diameter (B)
(C) (D)

Ans : 1 — Diameter [FSO 2020]

Q.11 Pressler's increment borer is used to determine the _____ of trees.

- (A) Age & Diameter Increment (B)
(C) (D)

Ans : 2 — Age & Diameter Increment [FSO 2020]

Basics of forest engineering and surveying

Q.12 Surveying can be regarded as the art and science of

- (A) Map making (B) Wood measurement
(C) Soil analysis (D) Forest inventory

Ans : (A) [FSO 2026]

Q.13 How many links are present in a 20 m long metric chain?

- (A) 50 (B) 200
(C) 100 (D) 20

Ans : (C) [FSO 2026]

Q.14 Mensuration is derived from the _____ word "Mensura" meaning measure.

- (A) Latin (B)
(C) (D)

Ans : 3 — Latin [FSO 2020]

15. REMOTE SENSING & SATELLITE COMMUNICATION

Forest cover monitoring through remote sensing

Q.1 In Remote Sensing Hyper Spectral Sensors are useful for

- (A) To discriminate one vegetation type to the another
- (B) To discriminate one plant to the another
- (C) To extract vegetation nonbiophysical information
- (D) To extract vegetation physical information

Ans : (A) [FSO 2026]

Q.2 Remote sensing is the science of obtaining information about an object without

- (A) Measuring it
- (B) Contacting it physically
- (C) Using satellites
- (D) Seeing it

Ans : (B) [FSO 2026]

Q.3 _____ is a mobile app which turns smartphone into a tool that tracks diseased trees and delivers forest health data to scientists and forest managers around the world.

- (A) Tree Tagger
- (B)
- (C)
- (D)

Ans : 3 — Tree Tagger [FSO 2020]

Geographic information Systems for management and modeling

Q.4 Temporal resolution refers to

- (A) Number of spectral bands
- (B) Size of pixel on sensor
- (C) Frequency of image capture
- (D) Band width of sensor

Ans : (C) [FSO 2026]

Usage of wireless sets & walky-talkies for communication and their basics

Q.5 Wireless communication in forests is most essential for

- (A) Tree plantation
- (B) Forest road construction
- (C) Fire alerts and poaching control
- (D) GIS digitization

Ans : (C) [FSO 2026]

16. FOREST PLANNING & MANAGEMENT PLANS**Forest planning, evaluation and monitoring tools and approaches for integrated planning**

Q.1 As per the National Working Plan Code - 2023, who shall head a Forest Working Plan ?

- (A) Forester
- (B) Conservator of Forests
- (C) Forest Guard
- (D) Deputy Conservator of Forests

Ans : (D) [FSO 2026]

Q.2 The Forest Management Information System (FMIS) integrates

- (A) Physical and human data
- (B) Policy and legal acts
- (C) Forestry data for planning and evaluation
- (D) Revenue and budgeting only

Ans : (C) [FSO 2026]

Q.3 What is main purpose of forest evaluation?

- (A) To clear forests for agriculture
- (B) To increase industrial development
- (C) To replace natural forests with plantations
- (D) To estimate the value and condition of forest resources

Ans : (D) [FSO 2026]

Q.4 Which one of the following is the most important criteria for determining the site quality of an area?

- (A) Top height of the Crop
- (B)
- (C)
- (D)

Ans : 2 — Top height of the Crop [FSO 2020]

Q.5 Life table refers to one of the basic tools of

- (A) Demographic analysis
- (B)
- (C)
- (D)

Ans : 3 — Demographic analysis [FSO 2020]

Q.6 _____ is a tabular statement showing the order and extent of all works to be carried out during the year.

- (A) Annual Plan of operations
- (B)
- (C)
- (D)

Ans : 3 — Annual Plan of operations [FSO 2020]

Q.7 _____ is a yield table in which parameters have been given only for one grade of thinning which is usually 'C' grade.

- (A) Single yield table
- (B)
- (C)
- (D)

Ans : 1 — Single yield table [FSO 2020]

Q.8 _____ is using a growth model to estimate the development of a stand through time under alternative conditions or cutting practices.

- (A) Simulation (B)
(C) (D)

Ans : 4 — Simulation [FSO 2020]

Working plans Versus Management plan

Q.9 _____ is defined as the time that elapses between successive main fellings in the same area

- (A) Felling cycle (B)
(C) (D)

Ans : 2 — Felling cycle [FSO 2020]

Sampling: need for sampling, basic concepts in sampling, designing large-scale sample surveys, sampling and non-sampling errors

Q.10 Systematic sampling involves selecting samples

- (A) At equal intervals (B) Randomly from clusters
(C) By judgement (D) Based on soil type

Ans : (A) [FSO 2026]

Q.11 Which of the following is a main assumption for using regression estimation?

- (A) No relationship exists between variables
(B) There is a linear and predictable relationship
(C) Only nominal data is used
(D)

Ans : (C) [FSO 2026]

Q.12 Multi-stage sampling is a method of _____, where sampling units are not taken out at one stage but are drawn from a population at multiple stages.

- (A) Random sampling (B)
(C) (D)

Ans : 4 — Random sampling [FSO 2020]

Q.13 _____ sampling is a method of sampling whose characteristic feature is that the number of observations in the sample is not determined in advance but sampling units are taken successively from a population.

- (A) Sequential (B)
(C) (D)

Ans : 4 — Sequential [FSO 2020]

17. ENVIRONMENTAL SCIENCE

Definition, scope and importance of environmental science and the need for public awareness on it

Q.1 The National Air Quality Index (AQI) was launched in

- (A) 2012 (B) 2013
(C) 2014 (D) 2015

Ans : (C) [FSO 2026]

Q.2 Which of the following is the basic definition of a brick?

- (A) A natural stone block used for building
(B) An artificial block of burnt clay used for building purposes
(C) A cement block used in construction
(D) A sun-dried block made of sand and lime

Ans : (B) [FSO 2026]

Q.3 The portion of the road used by vehicles is called

- (A) Berm (B) Sub-base
(C) Carriage way (D) Wearing course

Ans : (C) [FSO 2026]

Q.4 The need for public awareness of environmental science arises mainly due to

- (A) Reduced scientific research
(B) Rapid industrialization and population growth
(C) Excess agricultural production
(D) Limitation of environmental policies

Ans : (B) [FSO 2026]

Q.5 Earth day is celebrated every year on

- (A) April, 22nd (B)
(C) (D)

Ans : 5 — April, 22nd [FSO 2020]

Q.6 The National Green Tribunal Act was enacted in the year

- (A) 2010 (B)
(C) (D)

Ans : 3 — 2010 [FSO 2020]

Q.7 According to the Malin Falkenmark Water Stress Index, when fresh water availability is 1,000 - 1,700 cum per person per year, countries experience

- (A) Water stress (B)
(C) (D)

Ans : 1 — Water stress [FSO 2020]

Q.8 Disaster management Act was enacted in the year

- (A) 2005 (B)
(C) (D)

Ans : 3 — 2005 [FSO 2020]

Q.9 Roads are required to be _____ when the intensity of traffic passing over it rises

- (A) Metalled (B)
(C) (D)

Ans : 4 — Metalled [FSO 2020]

Q.10 _____ is a method of repairing irregular wear of the surface of the road.

- (A) Patching (B)
(C) (D)

Ans : 2 — Patching [FSO 2020]

Q.11 Culverts are small bridges with a span of less than _____ meters.

- (A) 4 (B)
(C) (D)

Ans : 4 — 4 [FSO 2020]

Q.12 _____ cultivation is also known as 'slash and burn' or 'Swidden cultivation'.

- (A) Shifting (B)
(C) (D)

Ans : 1 — Shifting [FSO 2020]

Q.13 _____ mechanically breaks up the compacted soil layers.

- (A) Deep Ripping (B)
(C) (D)

Ans : 4 — Deep Ripping [FSO 2020]

Q.14is most essential element for all life forms

- (A) Water (B)
(C) (D)

Ans : 4 — Water [FSO 2020]

Definition, Causes, effects and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution & Nuclear hazards

Q.15 Which gas is a major contributor to air pollution from vehicles?

- (A) Oxygen (B) Nitrogen
(C) Carbon monoxide (D) Helium

Ans : (C) [FSO 2026]

Q.16 Which pollutant is most associated with acid rain ?

- (A) Methane (B) Sulphur dioxide
(C) Carbon dioxide (D) Ozone

Ans : (B) [FSO 2026]

Q.17 Chronic exposure to low-frequency industrial noise primarily affects which human system?

- (A) Digestive system
(B) Skeletal system
(C) Nervous and cardiovascular system
(D) Endocrine system

Ans : (C) [FSO 2026]

Q.18 Which of the following is an effect of long-term exposure to noise pollution?

- (A) Reduced blood pressure
(B) Improved sleep quality
(C) Noise-induced hearing loss due to inner ear damage
(D) Strengthened nervous system

Ans : (C) [FSO 2026]

Q.19 Which of the following is a major source of thermal pollution?

- (A) Paper recycling units (B) Coal-fired power plants
(C) Wind energy plants (D) Solar power panels

Ans : (B) [FSO 2026]

Q.20 Any offence under Biological Diversity Act, 2002 is

- (A) Cognizable and non-bailable
(B)
(C)
(D)

Ans : 3 — Cognizable and non-bailable [FSO 2020]

Q.21 Reddish brown flecks on leaves, bleaching , premature aging, in plants is caused by

- (A) Ozone (B)
(C) (D)

Ans : 4 — Ozone [FSO 2020]

Q.22 Which of the following instrument is used for measuring the direction of the wind?

- (A) Weather-cock (B)
(C) (D)

Ans : 1 — Weather-cock [FSO 2020]

Solid waste Management: Causes, effects and control measures of urban and industrial wastes

Q.23 Composting is a solid waste management method mainly useful for

- (A) Non-biodegradable waste
- (B) Biodegradable organic waste
- (C) Metallic waste
- (D) Nuclear waste

Ans : (B) [FSO 2026]

Q.24 Bagasse is the fibrous residue left after extracting juice from

- (A) Wheat (B) Sugarcane
- (C) Rice (D) Jute

Ans : (B) [FSO 2026]

Q.25 Hazardous waste (Management & handling) rules came into existence in the year

- (A) 1989 (B)
- (C) (D)

Ans : 3 — 1989 [FSO 2020]

Q.26 The principle involved using activated carbon for wastewater treatment is

- (A) Adsorption (B)
- (C) (D)

Ans : 3 — Adsorption [FSO 2020]

Environmental ethics: Issues and possible solutions, Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust, Wasteland reclamation, Consumerism and waste products

Q.27 Wasteland reclamation involves

- (A) Converting forests into urban areas
- (B) Restoring degraded lands for productive use
- (C) Mining minerals from barren lands
- (D) Increasing industrial pollution

Ans : (B) [FSO 2026]

Q.28 A river basin project diverts water for intensive irrigation upstream, leaving downstream villages dry. Which principle of sustainable water ethics is violated ?

- (A) Maximization of yield for immediate gain
- (B) Privatization of resource rights
- (C) Equitable distribution and inter-regional justice
- (D) Precautionary principle

Ans : (C) [FSO 2026]

Q.29 What is the main purpose of Environmental Impact Assessment (EIA)?

- (A) To increase industrial output
- (B) To predict environmental consequences of proposed projects
- (C) To promote deforestation
- (D) To assess economic returns only

Ans : (B) [FSO 2026]

Q.30 Which process is responsible for trapping heat in the earth's atmosphere leading to global warming?

- (A) Ozone formation (B) Acid rain
- (C) Greenhouse effect (D) Photosynthesis

Ans : (C) [FSO 2026]

Q.31 Diversion of large forest lands was required for _____ project across Godavari River in Andhra Pradesh.

- (A) Polavaram (B)
- (C) (D)

Ans : 1 — Polavaram [FSO 2020]